

root of polynomial.



Def 1: Let $f(x), g(x) \in k[x]$.

We say $f(x)$ divides $g(x)$ if

$$\exists h(x) \in k[x] \text{ s.t. } h(x)f(x) = g(x)$$

We say $f(x) \in k[x]$ is irreducible

if $f(x) \neq 0$ is not a unit

and whenever $f(x) = g(x)h(x)$

for $g(x), h(x) \in k[x]$, $\deg g(x) = 0$

or $\deg h(x) = 0$,

eg. $x^2 + 1$ is irreducible in $\mathbb{R}[x]$.

factors as $(x+i)(x-i)$ in $\mathbb{C}[x]$

Def 2: Let R, S be commutative rings

A function $f: R \rightarrow S$ is a ring homomorphism

if $f(1_R) = 1_S$

$$f(a+b) = f(a) + f(b)$$

$$f(ab) = f(a)f(b) \quad \forall a, b \in R.$$

eg. Let K be a field and $p \in K$.

Define $ev_p: K[x] \rightarrow K$

$$\sum_{i=0}^n a_i x^i \mapsto f(x) \mapsto f(p) = \sum_{i=0}^n a_i p^i$$

This is a ring homomorphism called

evaluation at p .

def 3: let $p \in K$. We say p is a

root of $f(x) = \sum_{i=0}^n a_i x^i$ in K if

$$\text{ev}_p(f) = f(p) = 0 \text{ in } K.$$

Cor 4: let $p \in K$ and $f(x) \in K[x]$.

Then. p is a root of $f(x)$

$$\text{iff } x-p \mid f(x).$$

proof: ($\Leftarrow$) If $f(x) = (x-p)g(x)$

for some $g(x) \in K[x]$, then

$$\text{ev}_p(f) = \text{ev}_p(x-p) \text{ev}_p(g) = 0.$$

($\Rightarrow$) , Write $f(x) = (x-p)g(x) + r(x)$

where $r(x) = 0$ or $\deg r(x) < \deg x-p = 1$

In particular, $r(x)$ is a constant polynomial.

Since p is a root, $ev_p(f) = ev_p(x-p) ev_p(g) + ev_p(r)$

so $0 = ev_p(f) = ev_p(r) = r$. Then $f(x) = (x-p)g(x)$.

eg. $x^3 - 2x + 1$. By apply the rational

roots test and computing, one may

notice $x=1$ is a root.

Then we factor $x^3 - 2x + 1 = (x-1)(a_2x^2 + a_1x + a_0)$

By expanding the product and

comparing the coefficients.

$$a_2 = 1, \quad -a_0 = 1$$

$$-a_2 + a_1 = 0.$$

$$\text{So } x^3 - 2x + 1 = (x-1)(x^2 + x - 1)$$

Or one can observe that

$$x^3 - x^2 + x^2 - 2x + 1$$

$$= x^2(x-1) + (x-1)^2 = (x-1)(x^2 + x - 1).$$

Lemma 5: Any linear polynomial. $ax+b$

for $a \neq 0$ is irreducible in $K[x]$.

proof: Suppose $ax+b = g(x)h(x)$.

Then $\deg g + \deg h = \deg ax+b = 1$

So at least one of $\deg g, \deg h = 0$

that is. at least one of g or h

is a unit.

Corollary 6: let $f(x) \in K[x]$ be of $\deg n \geq 0$.

Then, f has at most n distinct roots.

proof. Suppose $f(x)$ has roots in K .

(If f has no roots, the statement holds true.)

Say $p_1, \dots, p_m$ are distinct roots

for some $m \geq 1$. Then $x - p_i$ are

distinct irreducible factors of $f(x)$

As $K[x]$ is ED which is a UFD.

we must write $f(x) = (x - p_1) \cdots (x - p_m) g_1(x) \cdots g_t(x)$

for some irreducible $g_j(x) \in K[x]$.

In particular $0 \leq n = \deg f(x) = m + \deg g_1 + \cdots + \deg g_t$

If $m > n$, this is impossible. So $m \leq n$.

eg. $\mathbb{Z}_2$ is a field. In $\mathbb{Z}_2[x]$.

the linear polynomials are irreducible

$$x, \quad x+1.$$

the quadratic polynomial if reducible, must

factor out a linear term, which are.

$$x^2, \quad x(x+1) = x^2 + x, \quad (x+1)^2 = x^2 + 1$$

We are left with one. $x^2 + x + 1$

Note. both 0 and 1 are not roots in $\mathbb{Z}_2$.

the cubic polynomial if reducible must factor

out a linear term. i.e. it must have a root.

so we first list all cubics. (8 in total)

$$x^3 \rightarrow 0 \text{ is a root}$$

$$x^3 + 1 \rightarrow 1 \text{ is a root}$$

$$x^3 + x \rightarrow 1 \text{ is a root}$$

$$x^3 + x + 1 \rightarrow \text{No roots} \rightarrow \text{irreducible}$$

$$x^3 + x^2 \rightarrow 1 \text{ is a root}$$

$$x^3 + x^2 + 1 \rightarrow \text{no roots} \rightarrow \text{irreducible}$$

$$x^3 + x^2 + x \rightarrow 0 \text{ is a root}$$

$$x^3 + x^2 + x + 1 \rightarrow 1 \text{ is a root}$$

